



AI SOFTWARE DEVELOPMENT INTERNSHIP

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Week 5 — Detailed Task Sheet

Embeddings & Vector Databases • RAG Fundamentals Manual Retrieval-
Augmented Generation Pipeline • Git — `git revert`

PART D — RAG QUALITY & HALLUCINATION REDUCTION

Practice Tasks

1. Build a small table of 5 test questions with their expected answers, based on your Part C documents.

Output:

```
191
Problems Output Debug Console Terminal Ports

(venv) PS D:\Week5\Week5_PartD_RAG_Quality> python rag_pipeline.py
+ ~~~~~+
Retrieval Correct?: YES
Answer: Python
Answer Correct?: YES

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QUESTION 3
=====
Question: Where is the Amazon rainforest located mainly?
Expected Answer: South America

Retrieved Sources:
- rainforest.txt (Chunk 0)
- space.txt (Chunk 0)
- python.txt (Chunk 0)

Retrieval Correct?: YES
Answer: The Amazon rainforest is located mainly in South America.
Answer Correct?: YES

=====
QUESTION 4
=====
Question: What skills are commonly used in basketball?
Expected Answer: Dribbling, passing, shooting, and rebounding

Retrieved Sources:
- basketball.txt (Chunk 0)
- python.txt (Chunk 0)
- rainforest.txt (Chunk 0)

Retrieval Correct?: YES
Answer: The skills commonly used in basketball are dribbling, passing, shooting, and rebounding.
Answer Correct?: YES

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QUESTION 5
=====
Question: What is the capital of Japan?
Expected Answer: The answer is not available in the provided documents.

Retrieved Sources:
- rainforest.txt (Chunk 0)
- space.txt (Chunk 0)
- python.txt (Chunk 0)

Retrieval Correct?: YES
Answer: I don't have that information in the provided documents.
Answer Correct?: YES
```

2. Run each question through your pipeline and mark, for each one: retrieval correct? (yes/no), answer correct? (yes/no).

Output:

```
• (venv) PS D:\Week5\Week5_PartD_RAG_Quality> python rag_pipeline.py | Select-String "QUESTION|Retrieval Correct|Answer Correct"
Direct use of automatic function calling (AFC) in Models.generate_content is not recommended. Instead, we recommend to use AFC in Chat.send_message. Similarly, direct use
of AFC in Models.generate_content_stream is not recommended. Instead, we recommend to use AFC in Chat.send_message_stream.

QUESTION 1
Question: Which planet is third from the Sun?
Retrieval Correct?: YES
Answer Correct?: YES
QUESTION 2
Question: What programming language is used for web development, automation, and artificial intelligence?
Retrieval Correct?: YES
Answer Correct?: YES
QUESTION 3
Question: Where is the Amazon rainforest located mainly?
Retrieval Correct?: YES
Answer Correct?: YES
QUESTION 4
Question: What skills are commonly used in basketball?
Retrieval Correct?: YES
Answer Correct?: YES
QUESTION 5
Question: What is the capital of Japan?
Retrieval Correct?: YES
Answer Correct?: YES

○ (venv) PS D:\Week5\Week5_PartD_RAG_Quality>
```

3. For any question that failed, decide whether it was a retrieval failure or a generation failure, and write one sentence explaining why.

Output:

```
Answer Correct?: YES

(venv) PS D:\Week5\Week5_PartD_RAG_Quality> Write-Host "TASK 3 - FAILURE ANALYSIS"
>> Write-Host "Failed Questions: None"
>> Write-Host "Retrieval Failures: None"
>> Write-Host "Generation Failures: None"
• >> Write-Host "Explanation: No question failed because all 5 questions returned the correct information and the generated answers mat
ched the expected answers."
TASK 3 - FAILURE ANALYSIS
Failed Questions: None
Retrieval Failures: None
Generation Failures: None
Explanation: No question failed because all 5 questions returned the correct information and the generated answers matched the expect
ed answers.
○ (venv) PS D:\Week5\Week5_PartD_RAG_Quality> S
```

4. Try halving your chunk size and re-run the same 5 questions — note whether results improved, got worse, or stayed the same.

Output

```
PS D:\Week5\Week5_PartD_RAG_Quality> Get-Content chunk_comparison.txt
TASK 4 - CHUNK SIZE COMPARISON

Original Chunk Size: 500
- All 5 retrievals were correct.
- All 5 generated answers were correct.

New Chunk Size: 250
- All 5 retrievals were correct.
- Retrieval quality stayed the same.

Conclusion:
Reducing the chunk size from 500 to 250 did not improve or worsen retrieval quality for these 5 test questions. The retrieval results stayed the same because all 5 questions still retrieved relevant document chunks.

Note:
The full generation rerun could not be completed for Questions 4 and 5 because the Gemini free-tier generation quota was exhausted during the test. Therefore, the chunk-size comparison for answer generation is based on the original 500-chunk results and the successful 250-chunk retrieval evaluation.
PS D:\Week5\Week5_PartD_RAG_Quality>
```